

Sonderdruck aus

UGARIT-FORSCHUNGEN

Internationales Jahrbuch
für die Altertumskunde Syrien-Palästinas

begründet von Manfred Dietrich und Oswald Loretz †

unter Mitarbeit von Josef Tropper

herausgegeben von
Manfred Dietrich und Ingo Kottsieper

Beratergremium
J. Bretschneider • K. A. Metzler
R. Schmitt • W. H. van Soldt • J.-P. Vita

Band 46
2015



Manfried Dietrich : ugarit@uni-muenster.de
Ingo Kottsieper : Ingo.Kottsieper@mail.uni-goettingen.de
Josef Tropper : josef.tropper@t-online.de

Redaktion

Ugarit-Verlag, Rosenstr. 9, D-48143 Münster
(Kai A. Metzler: metzler@ugarit-verlag.de)

Für unverlangt eingesandte Manuskripte kann keine Gewähr übernommen werden.

Die Herausgeber sind nicht verpflichtet,
unangeforderte Rezensionsexemplare zu besprechen.
Manuskripte für die einzelnen Jahressbände werden jeweils
bis zum 31. 12. des vorausgehenden Jahres erbeten.

© 2015 Ugarit-Verlag, Münster
(www.ugarit-verlag.com)

Alle Rechte vorbehalten

All rights reserved. No part of this publication may be reproduced,
stored in a retrieval system, or transmitted, in any form or by any means,
electronic, mechanical, photo-copying, recording, or otherwise,
without the prior permission of the publisher.

Printed in Germany

ISBN 978-3-86835-181-1

Printed on acid-free paper

Inhalt

Artikel

Joachim Bretschneider / Athanasia Kanta / Jan Driessen

Pyla-Kokkinokremos. Preliminary Report on the 2014 Excavations.....1

Guy Bunnens

On Upper and Lower Aram Again39

Collin Cornell

A Moratorium on God-Mergers? The Case of El and Milkom in the
Ammonite Onomasticon.....49

Manfried Dietrich

Die keilalphabetische Inschrift KTU 6.1 auf dem Bronzemesser
aus dem Tabor-Tal 101

Gershon Galil

A New Look at the History of Jabesh-Gilead (Tell Abu al-Kharaz)
in the Light of New Archaeological and Epigraphic Data.....105

Erasmus Gaß

Schoschenq und Jerusalem. Probleme einer historischen Rekonstruktion 115

Philippe Guillaume

Naboth the Nabob. A View from Assyrian Jezreel161

Issam K. H. Halayqa

Two Canaanite Scarabs from Kherbit el-Jami' (el-'Ubayat)
South of Bethlehem183

Kyle H. Keimer

The Impact of Ductus on Script Form and Development
in Monumental Northwest Semitic Inscriptions189

Joseph Livni

Investigation of Population Growth of Ancient Israel.....213

Valérie Matoïan

Horon et Shed à Ugarit: textes et images235

Ianir Milevski / Omry Barzilai / Nimrod Getzov / Ariel Vered

An Animal Relief from 'Ein Zippori, Lower Galilee289

Juan OlivaOn the Order of Shuppiluliuma in Syria: Reading the Hurrian of
the Qatna Letter TT1299**Julie Patrier**Détail et alimentation dans l'iconographie syro-anatolienne
(III^e–I^{er} millénaires av. J.-C.). Partie II: Pomme de pin ou gigot ?321**Christa Schäfer-Lichtenberger**

PTGYH – Divine Anonyma? The Goddess of the Ekron Inscription341

Ola WikanderEmphatics, Sibilants and Interdentals in Hebrew and Ugaritic.
An Interlocking Model373**Nicolas Wyatt**The Evidence of the Colophons in the Assessment of Ilmilku's
Scribal and Authorial Role399**Jonathan Yogev / Shamir Yona**

Visual Poetry in KTU 1.2447

Irit Ziffer / Dina Shalem"Receive my breast and suck from it, that you may live." Towards
the imagery of two ossuaries from the Chalcolithic Peqi'in Cave455**Buchbesprechungen und Buchanzeigen**Peter ALTMANN / Janling FU (Hrsg.): *Feasting in the Archaeology
and Texts of the Bible and the Ancient Near East* (Ludger Hiepel)489Michael PIETSCH: *Die Kultreform Josias. Studien zur Religionsgeschichte
Israels in der späten Königszeit* (Wolfgang Zwickel)495Anson F. RAINEY: *The El-Amarna Correspondence. A New Edition of the
Cuneiform Letters from the Site of El-Amarna based on Collations of
all Extant Tablets. Collated, Transcribed, and Translated by Anson F.
Rainey Z"L* (Josef Tropper)502**Abkürzungsverzeichnis509****Indizes**

A Stellen515

B Wörter516

C Namen	517
D Sachen	520

Anschriften der Mitarbeiterinnen und Mitarbeiter	525
---	-----

Emphatics, Sibilants and Interdentals in Hebrew and Ugaritic

An Interlocking Model

Ola Wikander, Lund

Basic points of departure	374
The historical pronunciation of the emphatic consonants.....	375
The emphatic sibilant/affricate	384
“Backed” emphatics as a feature spreading in stages	387
Interdentals and sibilants in Ugaritic and Hebrew	388
Possible conclusions from the interlocking developments of emphatics, sibilant and “quasi-sibilant” sounds in Northwest Semitic.....	391
Acknowledgment	395
References	395

This article deals with two oft-discussed and bewildering questions of ancient Northwest Semitic phonetics and phonology: that of the relationship between the sibilant and interdental phonemes and that of the nature of the emphatic sounds. It is my view that these two questions may influence each other in certain ways, and it is my intention to present a possible model for the development of the Northwest Semitic phonemes in question.

The two problem spheres are among the most vexing in Semitic phonological history generally. The first concerns the emphatic sounds: were they in some way “backed”, i. e. pharyngealized, velarized or uvularized (as in Arabic or many dialects of Neo-Aramaic) or glottalized/ejective (as in Ethiopic and often in Modern South Arabian)? Today, the trend is generally to view the glottalized/ejective pronunciation as the original one and the pharyngealized/velarized one as secondary (or even as originally confined to Arabic).

The question of the nature of the various sibilant and quasi-sibilant phonemes is even more complex. Here, I will concentrate on the developments of these various sounds in Northwest Semitic – specifically in Hebrew and Ugaritic, and to a certain extent Aramaic – though comparative evidence from other languages will sometimes be adduced as well.

Basic points of departure

If one is to make any sense of these confusing and interlocking developments, the following facts need to be kept in mind.

- The interdental series of consonants (*t̪*, *d̪*, *z̪*),¹ while undoubtedly present in early Northwest Semitic, was highly unstable in all its branches – including Ugaritic. The existence of separate graphemes expressing this series in Ugaritic must not blind one from noticing this tendency. A similar instability affected the lateral fricatives (*ʃ* and its emphatic counterpart *ʃ̣*),² which disappeared in many languages and often fell together with other phonemes.
- In transcriptions of foreign languages into Ugaritic, sibilants are often rendered as *t̪*, not as *s* or *š*. In Hurrian words/names, the corresponding voiced allophone is often rendered by *d̪* rather than *z*.³
- Regardless of the recent trend of regarding the pharyngealized/velarized pronunciation of the Northwest Semitic emphatics as a late phenomenon (possibly influenced by Arabic) and the glottalized pronunciation as the actual ancient pronunciation, the evidence for pharyngealization is not to be ignored (as I shall argue below). It is, however, quite clear that glottalization is still the oldest phonetic realization of the emphatics, and both these facts must be remembered. As we shall see below, the question of the realization of the emphatic sounds may also affect the history of the interdental fricatives, and vice versa.
- When discussing the question of the pronunciation of the emphatic sounds, it is necessary to be aware that, cross-linguistically, “backing” (pharyngealization or velarization) tends to spread throughout a lexeme, whereas glottalized/ejective sounds often tend to dissimilate from each other (i. e. the exact opposite development). Thus, the two possible pronunciations of emphatics in ancient Semitic languages tend to behave in diametrically opposed ways, which is often regarded as significant.⁴

¹ For typographical reasons (and because it is normal in Ugaritological circles), I here use the common (though strictly incorrect) transcription *z̪* for the emphatic interdental fricative. This transcription is derived from the value of the cognate sound in modern Arabic. For typographical simplicity, I write [ts] etc. for affricates. I use a superscript ʔ for glottalized/ejective consonants.

² Again due to the influence of Arabic on earlier comparative Semitics, the sound *ʃ̣* is sometimes rendered *d̪* (it is pronounced as a pharyngealized/velarized *d̪* in standardized Arabic). It was in all probability unvoiced in antiquity; it appears generally to have been realized as a voiceless emphatic lateral fricative in old Semitic languages.

³ See, e. g. Tropper, 2012, 111–112, 117–118.

⁴ However, there are exceptions or intermediate possibilities, such as in certain cases in Modern South Arabian, where a general laryngealization or “creaky voice” realization of emphatics can spread over an entire word in the Qishn dialect of Mehri (see Simeone-

- The sibilant series *s/z/š* was in all probability a series of affricates in early Northwest Semitic, only later changing to a pure sibilant realization.⁵ This fact has far-reaching consequences.
- This affricated pronunciation may have been preserved in some reading traditions in the case of Hebrew *šādhē* – the Ashkenazic (and Modern Hebrew) realization of the letter as [ts] need not necessarily be ascribed to Yiddish influence: why, in this were the only deciding factor, would the identification happen to this exact letter? Also, there is ancient transcriptional evidence for an affricate *š* in Northwest Semitic in the forms of Hebrew *špn* being transcribed into Demotic Egyptian as *tspn* and of *hšr* being rendered in Greek as αστρηρ (there are a number of other similar cases).⁶

The historical pronunciation of the emphatic consonants

In older grammars and textbooks of Biblical Hebrew, it is commonly stated that the letters *ṭēth* and *šādhē* were originally pronounced with pharyngealization or velarization, and that the third Hebrew emphatic, *qōph*, was pronounced as an uvular [q] – all three being pronunciations well-known from Arabic. Although, sadly, few “Western” students of Biblical Hebrew have tended to make the effort to pronounce the sounds in this way, there was at one time something of a consensus that this was the historical realization once represented by these letters. In later years, however, there has been a slow but clear shift of the pendulum towards the view that in Ancient Hebrew (and in Northwest Semitic generally), the emphatic letters were rather pronounced as glottalized ejectives, in the manner represented by many of the modern South Semitic languages. The movement in this direction may well be viewed as part of a more general (and in itself highly commendable) tendency not to view Classical Arabic as the “perfect Semitic language” in the way that seems to have been *a là mode* say, fifty years ago.⁷

Senelle, 1997, 383). However, as long as one is discussing glottalization/ejectivity of specific consonants in general as opposed to “backing”, the principle above is sound.

⁵ This view is today so common as to be regarded as standard (see, for example, Kogan, 2011, 61, who talks of a “growing consensus”, with ample references in the sections that follow). See also Kogan, 2011, 69.

⁶ Lipiński, 1997, 125. More transcriptional indications of this sort can be found throughout Steiner, 1982, the classic study on the subject.

⁷ The view that Hebrew emphatics were glottalized/ejective can now also be found in beginning textbooks – see e.g. Hackett, 2010, which teaches it. One should note, however, that there are modern studies that posit pharyngealization for all of Northwest Semitic (see, e.g., Zelmánek, 1996, 4).

Arguments for a “backed” pronunciation of emphatics in Northwest Semitic

There are, however, a number of arguments suggestive of a pharyngealized or velarized (or uvularized) pronunciation of (some of) the emphatics in ancient Northwest Semitic. Of these, the most apparent and intuitive is also the most unconvincing when subjected to thorough scrutiny, viz. the tradition of pronouncing them in this manner in reading traditions of Syriac and Mizrahi or Yemenite Hebrew. This fact, while seemingly indicative of an ancient pharyngealized/velarized feature, has been brushed aside by much modern scholarship based on the (on the face of it) rather plausible suggestion that all these traditions have come to be due to adstrate influence from Arabic. Thus, it has been suggested that this “backed” pronunciation of emphatics is indicative of nothing more than Arabic having been the spoken language in the regions where these traditions of Hebrew and Syriac arose and of the pronunciation of the emphatics in spoken Arabic having been superimposed on the phonological systems of the traditional, religious languages.⁸ This would certainly not be a unique phenomenon – one could for example note the accommodation of the Coptic language to the Arabic phonological system that took place during the Middle Ages. There is, however, still the tantalizing question of why the Arabic-speaking users of mediaeval Hebrew and Syriac would specifically have identified ejective phonemes with the pharyngealized/velarized phonemes of Arabic – of course, the apparent identity of the names of many of the letters may have helped if such a process really occurred.

There are, however, more compelling reasons for maintaining pharyngealization/velarization as a probable pronunciation of at least some of these ancient Northwest Semitic phonemes, or rather, for postulating that such a pronunciation was in development during the productive period of the classical Northwest Semitic languages, independently of Arabic. The important point is not really which of the two pronunciations of emphatics (glottalization/ejectivity or “backing”) is the older one – it is quite clear that glottalization is the more archaic pronunciation, which may well have persisted here and there in Northwest Semitic as well.⁹ The question is whether the “backed” pronunciation is only attributable

⁸ For an example of this view concerning Hebrew, see Rendsburg, 1997, 75–76.

⁹ There is one Jewish reading tradition that uses ejective emphatics, viz. the Georgian one. But, as pointed out by Lipiński (1997, 106), this pronunciation is historically suspect due to the replacement of the actual pharyngeal phoneme *ʿayin* by a glottalized velar (sometimes described as a glottalized uvular) in that tradition, which shows that it had a general tendency to replace “backed” phonemes with ejective ones. In the case of Georgian Hebrew, adstrate influence from Caucasian languages (which abound in ejectives) must not be ignored. There is also some evidence for glottalization in certain Neo-Aramaic dialects (see Dolgopolsky, 1977, 1), which (given the perspective argued here) could well be viewed as an archaism (most Neo-Aramaic dialects do not use glottaliza-

to Arabic influence or rather, as I will argue, was developing across the North-west Semitic area even in antiquity. I will here point out a number of reasons for such a view, some of which have often been adduced in the literature, while others are, as far as I know, more specific to the present study.

- 1 The evidence of historical emphatics being rendered as velar or uvular phonemes in Ugaritic and early Aramaic.
- 2 The evidence of hithpa^ʿel forms in Hebrew, and parallel phenomena in Aramaic.¹⁰
- 3 The vocalization of the Hebrew word *qoṭobkā* (and parallels, such as *qorobkem*).
- 4 The lack of spontaneously appearing emphatics based on vowel-less sequences of Hebrew (or Aramaic) *s*, *t* or *k* and a following glottal stop.
- 5 The fact that Aramaic initially showed dissimilation of two emphatics in the same root (during the Old Aramaic period) but later displays no such process.

I shall now discuss the first four of these arguments in detail; the fifth will be taken up at the end of the article, after a digression on sibilant and interdental phonemes.

Argument number 1: backed graphemic reflexes in Ugaritic and Aramaic

Point 1 in this list refers to the fact that (under certain, not entirely, clear circumstances) the etymological emphatic interdental fricative (often, though misleadingly, transcribed *ẓ) appears in Ugaritic writing using the sign normally used to indicate the voiced uvular fricative, ġ. Similarly, early Aramaic renders the etymological emphatic lateral fricative sound *š (also sometimes transcribed ḏ due to its reflex in Arabic) using *q*,¹¹ and later Aramaic uses *ʿayin* for the same purpose.¹² The *q* rendering is less than conclusive, as that letter is of course also subject to the uncertainty concerning the realization of the emphatic feature

tion but “backing” or simple lack of aspiration to mark the “emphatic” feature). But note that Dolgopolsky points out (n. 2) that even few those dialects of Neo-Aramaic that *do* use glottalization also attest to a “backing” of vowels in the vicinity of emphatics, which points to those dialects being “on the fence” and in the process of a sound change. This is actually stated by Dolgopolsky himself on pp. 6–7.

¹⁰ These first two arguments are basically the same as some of those adduced in Lipiński, 1997, 105–106. Lipiński is of the view that pharyngealization/velarization was the *original* feature characteristic of the emphatic sounds of Semitic as a whole, a view which is clearly untenable in light of evidence from both Akkadian and South Semitic.

¹¹ A similar rendering appears to occur in the notoriously hard-to-identify dialect of the Deir Alla inscription as well (see Hackett, 1984, 111–112).

¹² Jer 10:11 famously shows both of these spellings in a single verse.

(creating a risk of circular reasoning).¹³ The Ugaritic *ḡ* and later Aramaic *ʿayin* are, however, more telling. As Lipiński argues, these are most easily explained by assuming that the emphatic feature inherent in the unstable interdental phoneme was some sort of co-articulation in the velar, uvular or pharyngeal region.¹⁴ An ejective/glottalized articulation would not provide as ready an explanation of this phenomenon.¹⁵ We will return to this matter later on.

Argument 2: emphasis spread in hithpaʿel forms

Point 2 is that of the Hebrew hithpaʿel forms such as *hiṣtaddēq*, which appear to show emphasis spread (*t* becoming *ṭ* in the vicinity of *ṣ*). This is easier to explain if *ṣ* was articulated in a way that included pharyngealization/velarization, as such articulation tends to spread throughout a word, whereas ejective phonemes would rather be expected to dissimilate in such an environment, yielding **hiṣtaddēq* with a *tāw*.¹⁶ When reading the literature, one sometimes gets the

¹³ Kaufman, 1997, 119 argues that the sound in Old Aramaic was actually articulated as “a velar spirant”, which would make the parallel with the Ugaritic development nearly perfect. For further, and more detailed, analysis of this Aramaic phoneme, see below (and footnote 44).

¹⁴ In his monumental grammar, Josef Tropper is cautiously positive towards the idea of a backed pronunciation of at least some of the emphatic sounds of Ugaritic (see Tropper, 2012, 98). He views the uvular-like pronunciation of the etymological interdental, sibilant and lateral emphatics as a sign in this direction (as do I). Note that he specifically mentions *ṣ* as probably still having been glottalized, however, a point of view much in line with what will be argued later in the present study.

¹⁵ There is another Afro-Asiatic typological parallel to the development of (something like) pharyngealized or velarized phonemes into an actual pharyngeal phoneme. This concerns the outcome of historical, Afro-Asiatic *d* (and possibly also *ḏ* and *z*) in Egyptian, a language that – according to the “new school” of Egyptian historical phonology – shifted this sound to *ʿayin*. This process is explained by Antonio Loprieno (1995, 31) as having probably come to pass through a stage in which the apical phonemes first turned into a lateral pronounced with pharyngealization, [ɬ]. The pharyngeal feature was allegedly then, so to speak, purified into an actual pharyngeal phoneme [ʕ]. Loprieno himself observes the parallel with the Northwest Semitic development on p. 244, n. 10. A similar typological comparison with Aramaic is done in Peust, 1999, 100. Peust also endorses the idea that Egyptian *ʿayin* was originally a dental which shifted in articulation into a pharyngeal around 2000 BC (p. 99). This analysis of the Egyptian *ʿayin* phoneme (which has many adherents today, although others regard it as controversial) originated in Rössler, 1971, esp. pages 275–278.

¹⁶ The tendency for ejectives to dissimilate is common cross-linguistically, and is represented in Semitic as well, for example in the Akkadian “Geers’ Law” (which is actually one the most important arguments for Akkadian emphatics having been ejectives; see Kogan, 2011, 60). The importance of the Hebrew *hiṣtaddēq*-type assimilation was elo-

impression that *hiṣṭaddēq* is the only word in Hebrew showing this development, but one should not discount examples such as *hiṣṭā'ēr* ("to be pained"), which occurs in Mishnaic Hebrew. A functionally identical phenomenon exists in Aramaic.

As mentioned, this assimilation is much easier to explain if some sort of backed coarticulation is involved (and downright strange if the phonemes are to be analyzed as glottalized ejectives) – *presupposing* that the writings actually reflect two emphatic sounds following each other and not an etymological "group writing" in which a glottalic release occurred only after the second sound (on this later problem, see below, footnote 29).

Argument 3: "backed" vocalism in Hebrew words with emphatics; words like *qoṭobkā*

Point 3 on the list is concerned with vocalism. A common criticism against interpretations of Hebrew emphatics as pharyngealized, velarized or uvularized is that the language does not appear to display "backing" of vowels close to these consonants. In languages that have "backed" emphatics, such "vowel backing" tends to be conspicuous – see, for example, the Arabic vowel *a*, which is mostly pronounced [æ] in neutral environments, but turns into a backed vowel [ɑ] when it occurs next to emphatics. The lack of such a process in Tiberian Hebrew seems telling, as the Masoretes were extremely thorough in recording the minutest differences in pronunciation.

There are, however, two arguments against this objection. The first is simply the fact that a "lighter" velarization would not necessarily have the big impact on vowels typical of full-blown pharyngealization, yet is still a form of "consonant backing".

The other (point 3 on the list) is that I actually believe there are isolated examples in which emphatic letters *do* change the vowels of a Hebrew word in a way reminiscent of that commonly occurring with the "guttural" letters. The first example is the suffixed nominal form *qoṭobkā*, which means "your destruction / your sting" (probably from **qōṭeb*) and occurs once, in Hos 13:14. According to the normal rules for **quṭl* segholates, the form ought to have been *qoṭbēkā*, with one *qāmeṣ hāṭūph* and not two. Another similar word is the suffixed infinitive *qorobkem* ("your drawing close") in Deut 20:2.¹⁷ Other similar examples are *qobollō* ("his battering-ram(?)", Hes 26:9) and possibly *qoṭonnī* ("my little

quently stated by Alicia Faber (1980, 140–141). Faber also points out the "dissimilatory nature" of ejectives (in relation to the Akkadian law of Geers) on pp. 145–147.

¹⁷ Joüon and Muraoka (2006, 166, § 65c) simply refer to this phenomenon as being meant to make pronunciation easier. In their section on nominal inflection (2006, 274, § 96j), Joüon and Muraoka just note the strange vocalism of the words and compare them to *po'olkā*, without further comment.

one” = “little finger” or “penis”, 1 Kings 12:10 etc.), although the gemination of the final consonant in these two words may make them less important for the present argument.

The interesting things to note here is (a) that all these words contain a *q*, and (b) that they appear to behave as if they contained a “guttural” – the insertion of a second *qāmeṣ ḥāṭūph* is quite typical of words such as *pōʿal*, which produces the suffixed form *poʿolkā*. The same phenomenon occurs in the word *ʾoholkā* (“your tent”), which also contains a “guttural”. That is: the words *qoṭobkā* and *qorobkem*, which contain both *q* and *t* in one case and *q* by itself in another use vowel patterns typical of similar roots containing a “guttural” (as do the other words mentioned above – note the dual emphatics of *qoṭonnî*). This, I would say, is an argument for a “backed” pronunciation of Hebrew emphatics in the classical era (at least of *q*).¹⁸ Note that the second case (*qorobkem*) – even though only including one emphatic and not two – does include an *r*, which regularly patterns with “gutturals” in the Tiberian phonological system (and also causes “backing” of vowels in Arabic). Of course, this argument is only applicable to the Tiberian Masoretic system (as opposed to ancient Hebrew generally), and the Tiberian vocalization is late and could theoretically have been influenced by Arabic, yet it is the Tiberian system that has been adduced as a point against “backed” emphatics in Hebrew, and forms such as *qoṭobkā* shows that emphatic letter *could* influence vowels in a way reminiscent of “back” consonants. Also, one should remember that in a “pure” transcription of these words in Tiberian vocalization (without “Qimḥian” vowel analysis), the words are *qāṭābkā* etc. (using *ā* for Tiberian *qāmeṣ*), clearly showing the backed “a/o-like character” of the vowels involved.

Argument 4: the lack of assimilation to an adjacent ʾāleph

The fourth point on the list is somewhat weaker, as it entails an *argumentum e silentio*; however, I believe that it may still be suggestive. My argument runs as follows: If the feature [EMPHATIC] were really fully equated with glottalization in the extant, historical Hebrew texts, one could well have expected that sequences of unemphatic *k*, *s* or *t* followed by a pronounced ʾāleph would regularly coalesce into the corresponding emphatic sound and grapheme (*q*, *ṣ*, *ṭ*). This would be likely, as an ejective is phonetically made up of a consonant made not through pulmonary pressure but through pressure from air above the closed glottis followed by a distinct “clicking” opening of the glottis (i. e., one

¹⁸ It is quite interesting to note that an early scholar such as F. E. König (1895, 68–69) noted these words, the general phenomenon and its connection with *q* and emphatics, without (of course) using that as an argument for the pronunciation of those letters. He expressly noted that these words have been given a “vollere Aussprache” due to the presence of emphatic consonants. Bauer and Leander (1965 [1922], 582) specifically mention that *qōṭeb* is inflected “wie bei einem Stamm med[iae] laryng[alis]”.

can somewhat inaccurately describe ejective consonants as being pronounced with an accompanying glottal stop). In fact, exactly this type of assimilation/conflation is attested in Assyrian, which has been used as a plausible argument that the emphatics of that language were at least usually articulated as ejectives.¹⁹ In Hebrew, however, nothing of the sort happens. One can look at words such as *pit'ōm* (“suddenly”), which is never written **piṭōm*, *mak'ōb* (“pain”), which does not turn into **maqōb* and, for that matter, forms of *t'm* (a root having to do with “twins”), which occurs in the Hebrew Bible in the form of the hiph'il participle *mat'imôt* (Song of Songs 4:2), a form that is certainly not turned into **mat'imôt*.²⁰ This root provides even more interesting evidence later on (in this case from Aramaic). I am thinking of the New Testament name (or epithet) Θωμᾶς, which is derived from Aramaic *t(ē)'ōmā'*. In this word (at least in the form underlying the Greek rendering), it appears that the *šēwā'* is not audibly present.²¹ This would create an eminent opportunity for the *tāw* and the *'āleph* to coalesce (theoretically into a *tēth*) if the emphatics were indeed generally pronounced as ejectives in the Hebrew and Aramaic of the New Testament period. However, the spelling of the Greek rendering strongly points in the other direction. The initial consonant is transcribed by Greek *thēta*, the aspirated dental stop. That an aspirated stop should also be ejective is extremely unlikely, and this means that the probability of Greek speakers rendering a putative ejective dental stop using *thēta* should be close to zero.²²

¹⁹ The word that has been invoked is *našā'u* (“to lift”, cognate with Hebrew *nāšā'*), which in Middle and Neo-Assyrian has given rise to forms written with *š* when the *š* meets the old glottal stop without an intervening vowel; this would point to Assyrian *š* having been glottalized/ejective (see Aro, 1977, 8). This line of reasoning was also accepted by Kouwenberg (2003) and Kogan (2011, 60). Kouwenberg added similar Old Assyrian (and occasionally later) evidence concerning the verbs *wašā'um* (“to go out”), *mašā'um* (“to be sufficient”), *kašā'um* (“to be cold”) and *nadā'um* (“to lay down”), using a somewhat different phonological analysis. I find the arguments in these cases to be quite compelling.

²⁰ In writing these theoretical (and nonexistent) forms, I have not made these words conform to Masoretic rules of vowels and syllabification. The reason that the example I am using here is not the verb *nāšā'* (in parallel with the Assyrian evidence presented in the previous footnote) is that Hebrew appears to have kept the ancient pronunciation of *š* as a lateral fricative for a significant time, which makes the historical development relative to the sibilants less clear.

²¹ Like in Ιούδας for *Y(ē)hūdā*.

²² One may compare with the system employed in the spelling of the Bohairic dialect of Coptic (also based on the Greek alphabet), in which what appears to be etymologically glottalized ejective stops are rendered using the voiceless unaspirated series (*kappa*, *pi* and *tau*), whereas etymologically non-ejective stops are rendered using aspirates (*thēta*, *phi*, *chi*) before a stressed vowel, semivowel or sonorant (see Loprieno, 1995, 42–44). In the Θωμᾶς case mentioned above, one could of course object that the origin of the Greek

Thus it has been shown that, in those cases where one could reasonably have expected an emphatic stop to appear in Hebrew and Aramaic because of language-internal phonological processes if the emphatics were consistently pronounced as ejectives, emphatic stops do not occur.²³ This is an argument somewhat lessening the probability of the Hebrew and Aramaic emphatics having been ejective in the classical period.²⁴

form may have been a variant that had completely lost the *ʾāleph* (such as is represented in the Syriac form, where the *ʾāleph* is only orthographic), but if the emphatics really were glottalized/ejective, such a form would also be an eminent candidate for the *t* and *ʾ* coalescing in to *t*, which, again, did not happen. Thus even this would be an argument in the same direction.

²³ As pointed out earlier, this argument entails an *argumentum e silentio*. As mentioned, Assyrian shows the type of development searched for here (which argues for that language having ejectives) – but Geʿez (a language which demonstrably has ejectives) does not. This shows that the absence of *ʾāleph* causing assimilation into ejectives is in itself not a definitive argument against ancient Hebrew having ejectives as a prominent part of its phonological system. However, if one presupposes that Hebrew emphatics *were* ejective, and that such assimilations as *št* in *hištaddēq* therefore represented not pharyngeal emphasis spread (as I argue, point 2 in the main list above) but “group writings” of a phonetic sequence [*stʔ*], in which the glottalic release occurred after the whole cluster (see the discussion of this possibility in footnote 29 below), one would have a situation in which the glottalic stop/release (which is, after all, phonetically identical to an *ʾāleph*) were subject to motion between and integration within other consonants, which would increase the likelihood of such an assimilation as the one searched for in vain above (e.g. *piṭʾōm* > **piṭōm*). All this rests, however, on a type of “contrafactual” linguistics which certainly has methodological difficulties attached to it.

²⁴ It is possible to make a similar argument, though even more uncertain, based on the Syriac writing of the name commonly rendered in a Greco-Latinized form as Paphnutius (a name of several different Coptic ecclesiastical figures). This name originally represents a Coptic form *Papnoute* (meaning “the one belonging to God”, with the second half of the word reflecting the ancient Egyptian word *nṯr*) and is rendered in Syriac as *Pafnūt*. What I find noteworthy here is the use of the Syriac emphatic *t* to signify what was in all probability an ordinary unvoiced, unaspirated dental stop in Coptic; ancient Coptic probably had glottalized ejectives as part of its phonetic inventory (see again Loprieno, 1995, 42–44 and note 22 above), but the *t* of *noute* was not one of these (as shown by its being descended from Egyptian *nṯr* which uses the unemphatic palatal stop transcribed as *t*). Thus, Syriac uses emphatic *t* to render nonejective Coptic *t*, which would be rather unexpected if a distinction nonejective-ejective were really present in Syriac Aramaic. Again, this would weaken the case for the Aramaic emphatics having necessarily been realized as glottalized ejectives in the classical period (and thus, by extension, somewhat strengthening the case for pharyngealization/velarization, or at least “something different than ejectivity”; of course a simple lack of aspiration could also be posited).

Against this line of reasoning two counterarguments could be adduced. The first one is that the Syriac *t* could simply have been intended to render the lack of aspiration in the original form or that the choice of the emphatic was made in order to avoid a pro-

Earlier in the history of the Hebrew language, however, such a process may actually have been in action. This is suggested by the co-occurrence of the roots *dqq* and *dkʿ*, both of which mean something like “to crush”. If these roots really are related (and I for one find that to be a highly likely proposition), it appears that the adjacent *ʾāleph* really did turn an unemphatic *k* into its emphatic counterpart once upon a time (or the other way around, meaning that *q* was segmented into *kʿ*).²⁵ This phonological process, however, apparently ceased to be freely productive before Hebrew comes into its historically attested phases. This, in itself, is an argument for the realization of the emphatic feature of Hebrew consonants having at least started to switch from glottalization to something else already during antiquity.²⁶

nunciation [θ], caused by Aramaic *bēghadḥkēfath* spirantization of *t*. The other consists in the possibility that the name *Pafnūt* was actually borrowed from the Greek form Παφνούτιος and not directly from Coptic (not having been “corrected” and influenced by the actual Coptic form), in which case, again, the choice of Syriac *t* could have been intended simply to convey lack of aspiration. There is also another Syriac form of the name ending in the letters -ys, which certainly represent the Greek ending -ιος. The possibility that Syriac used emphatic *t* simply to avoid spirantization into [θ] is rendered somewhat less probable by the fact that no such qualms stopped the Syriac transcription with what became (by spirantization) an *f* (but in that case one must remember that there is no emphatic *p* to use in Syriac, so the only recourse would have been an artificial gemination). All in all, though, I believe that the rendering *Pafnūt* somewhat lessens the likelihood for the Syriac emphatic having necessarily been pronounced as ejectives.

²⁵ For one version of the relationships, see Wilson-Wright, 2011. He also mentions the example *ʾāḥattennā* from Gen 31:39 (from the root *ḥtʿ*). This is certainly an interesting example, yet the root already contains a *t*, which makes it less definitive as a proof of the phonetic process still being “alive”. Also, the derivation of the form from that root is uncertain (see Loretz, 1975, and Loewenstamm, 1978).

²⁶ As an aside, one might also reflect on a matter of onomatopoeia. I am thinking here of the Hebrew word *baqbūq*, “flask”. As it stands, this word works perfectly as an onomatopoeic representation of the sound of liquid coming out of a flask – it actually works best if one pronounces *q* as a uvular stop. The interplay between the labial *b* and the uvular *q* really does sound like water spilling forth in bursts. If one treats *q* as a velar ejective stop, however, the effect becomes rather less impressive: [bakʷbu:kʷ] does not sound much like water coming from a flask. This word, then, could also point to a general “backing” of the emphatic phonemes in Hebrew. However, one should remember that the root in question seems originally to have existed in a “glottalized” version also having to do with such “watery” sounds: in Geʿez, there is a word *baqbaqa*, specifically in the derived form *ʾanbaqbaqa*, which means “cause to gargle, bubble”, which can be etymologically connected with the Hebrew “flask” word (see Leslau, 1987, 100 for the meaning and etymological relationship). Thus, the Hebrew word is not decisive; however, its formation is an unusually fitting onomatopoeic fit if the Hebrew *q* is interpreted as a uvular stop. If this is a historical fluke or indicative of something more is another question.

The emphatic sibilant/affricate

The possibility of *š* having kept ejectives for a longer time than other emphatics

Although I believe that the pharyngealized/velarized pronunciation of the emphatics had started to spread through Northwest Semitic already in antiquity, it was probably not ubiquitous. I would like to argue the possibility that while the “emphatic feature” had in Northwest Semitic sometimes switched from being phonetically represented as glottalization to something like pharyngealization or velarization, at least one phoneme, *š*, could still have kept glottalization as an additional phonetic trait for a longer time.²⁷ Such a state of affairs would account well for the phenomenon that there are a number of signs (mentioned above) of an affricated pronunciation of this phoneme (as opposed to unemphatic *s*) in later times.²⁸ It is known from the Ethiopian Semitic languages that an ejective *š* has a tendency to sound like an affricate. Thus, in this phoneme, Northwest Semitic may have kept glottalization as a coarticulation – or rather, affrication and glottalization may have helped preserve each other during the two processes where (1) the emphatics evolved from being glottalized to pharyngealized and (2) the affricates became deaffricated and transformed into normal sibilants. That is, both of these archaic (and, in Northwest Semitic, fading) features – affrication and glottalization – served to keep each other in existence.

A transitional *š* having both co-articulations at the same time?

Yet, spellings such as the above-mentioned *hišṭaddēq* (from the root *š-d-q*) suggest that there was a “backing” (pharyngealization/velarization/uvularization) as well – otherwise, the emphatic *š* would hardly turn the *t* into its emphatic counterpart (see above). Thus, I think it quite possible that (at least at an early stage of Northwest Semitic), the phoneme transcribed as *š* was phonetically something like [tsʔ], showing both pharyngealization and glottalization at the

²⁷ It is not impossible that the other emphatics were also in a state of flux between glottalization and backing (see the *ʾāḥattennâ* example from the footnote 25, for example, as well as the use of Hebrew emphatic letters to render emphatics in Babylonian personal names). I am talking of a gradual change during the lifetime of Biblical Hebrew. One should also keep in mind that we need not imagine that *all* emphatic phonemes in a given language were articulated in the same way: a modern example of a “mixed” situation can be found in Mehri, which uses pharyngealization for some of its emphatics and glottalization for some of them (see Watson, 2009, 6–10, with further references).

²⁸ For further, and deeper, analyses of *š* as an affricate, and a very thorough survey of the various reading traditions of Hebrew and Aramaic providing evidence for such a pronunciation, see Steiner, 1982. See also Kogan, 2011, 62–63.

same time.²⁹ This redundantly feature-prone phoneme would probably eventually shed glottalization in some dialects, turning into [tsʔ] (later Ashkenazic pronunciation, which shed all traces of a specific emphatic realization, naturally reinterpreted this as simple [ts], which has persisted even into Modern Israeli Hebrew). The retention of the affricate pronunciation was probably not universal, but restricted to certain dialects, as indicated by the varying reading traditions of Hebrew.

The existence of emphatic sounds both glottalized and pharyngealized at the same was also suggested by I. Maddieson.³⁰ Maddieson's suggestion was, however, concerned with the original, proto-language, pronunciation of the emphatics. The central idea would be that such a double coarticulation would explain the two attested pronunciations of various Semitic languages in a simple way. This is not the point I am making: rather, I am suggested that *ṣ* kept glottalization as a fossilized remnant, or better: *that the affricated and glottalized features helped preserve each other.*

The idea that *ṣ* kept its affricated pronunciation due to the influence of the ejective realization of the emphatic feature (in a way similar to what is found in

²⁹ As pointed out obliquely earlier, could possibly also analyze writings such as *hiṣtaddēq* as a sort of etymologizing “group writing”, with only the latter of the two sounds actually being realized as an emphatic. If that is so, one could equally well imagine a glottalized realization with the actual glottalic feature phonetically occurring after the second of the sounds: [histʔaddekʔ]. This type of explanation is used in Louwenberg, 2003, 84, n. 56 to account for similar spellings of perfect forms in Neo-Assyrian (*iqṭibi* for etymological *iqṭibi*, “he has said”), which have been taken to represent the rise of pharyngealization/velarization in Assyrian as well (so Huehnergard, 1997, 438). Howsoever that may be, I personally feel that such an explanation (that the writing does not really represent two pronounced emphatics) would be less compelling in the Hebrew case of *hiṣtaddēq*, etc.; given what is argued here concerning the pronunciation of Hebrew *ṣādhē* (accepting that it appears to have had an affricated pronunciation in antiquity, at least in some contexts and dialects), I find it somewhat hard to believe that the phonological process involved would produce a very complex cluster that would have sounded something like [tsʔ]. Rather, one would have expected a process of deaffrication of the (theoretical) glottalic dental affricate before a dental stop (cf. a parallel process in Ugaritic and Akkadian, see Kogan, 2001, 63), which would have produced *sāmekh-tēth* or possibly *šin-tēth*. If, on the other hand, pharyngealization/velarization had already started to appear in period of Classical Hebrew, and the letter *ṣādhē* was realized (as is argued above) something like [tsʕ] or [sʕ], it is quite easy to imagine the pharyngeal part of the coarticulation spreading to the dental stop as well, whereas the glottalic one could either just disappear or be pronounced after the cluster as well. Also, in such environments, one could well imagine that the affrication of *ṣādhē* would be neutralized – but not the pharyngealization/velarization, which would still keep the spelling with that letter relevant.

³⁰ Personal comment published in Edzard, 1998, 102.

the Ethiopian languages) has been proposed before, notably by Vilenčik, and then later by Richard C. Steiner.³¹ I would, however, like to extend the concept to the possibility that the influence “worked both ways”, so to speak. The evidence for pharyngealization/velarization/uvularization having come into being as a phonetic trait of the emphatics in Northwest Semitic is strong enough not to be ignored, and thus it cannot be sufficient to presuppose that glottalization was the universally used feature.

Ancient descriptions in St. Jerome

Also, one should not discount what St. Jerome says about the various sibilant letters of Hebrew:

*Siquidem apud Hebraeos tres s sunt litterae: una, quae dicitur samech, et simpliciter legitur quasi per s nostram litteram describatur: alia sin, in qua stridor quidam non nostri sermonis interstrepit: tertia sade, quam aures nostrae penitus reformidant.*³²

“Since indeed there are three *s* letters among the Hebrews: one, which is called *sāmekh*, and is simply read as what is indicated by means of our own letter *s*, another one called *šîn/śîn*, in which a certain hissing sound not occurring in our language can be heard, and a third one called *ṣādhē*, at which our ears deeply shudder.”

Though such arguments are perforce subjective in nature, I personally find it rather unlikely that Jerome would describe a simple glottalized *s* as so “scary” a sound. Ejectives are, after all, rather weak, and an ejective *s* would probably sound more or less like an ordinary *s* followed by a hiatus before the following vowel to a speaker of Latin – or possibly like a *ts* (which would hardly seem shudderworthy to Jerome) or as a slightly hissing *s*. If, however, we imagine that the sound that Jerome referred to was something like [tsʰ] or [sʰ], the “shuddering ears” of the Latinate saint become much more understandable.

Jerome also says the following:

*.... sade [...], cuius proprietatem et sonum inter z et s Latinus sermo non exprimit. Est enim stridulus, et strictis dentibus vix linguae impressione profertur.*³³

“... *ṣādhē*, the nature and sound of which, between *z* and *s*, the Latin language cannot express. For it is hissing/buzzing, and is produced with

³¹ Vilenčik, 1930, 92 (*non vidi*); Steiner, 1982, 85, 88–89. Steiner is, however, undecided as to whether the affrication is an original Proto-Semitic feature or rather a by-product of glottalization.

³² *Liber interpretationis Hebraicorum nominum* (CCSL 72), 71.

³³ *In Isaiam*, comment on Isa 11:1–2 (For the importance of this statement, see already König, 1881, 35).

difficulty, with the teeth “drawn together”/“tightened” with pressure of the tongue.”

This, again, sounds like quite a powerful sound; it could theoretically be interpreted as referring to an ejective – it all comes down to what is specifically meant by *stridulus* (“hissing/buzzing”) in this context. The reference to “difficult” and “pressure of the tongue” could well point to velarization or pharyngealization as well. One could theoretically also think of some sort of half-voiced phonation (similar to what is found in modern Mehri),³⁴ given Jerome’s talk of *ṣādhē* having a sound “between *z* and *s*”.

“Backed” emphatics as a feature spreading in stages

I would argue that the appearance of pharyngealization/velarization/uvularization was a phenomenon that spread as an areal feature across several Semitic languages, and did so by stages, and perhaps not always affecting all phonemes simultaneously or to the same degree.

For example, while most scholars would agree that glottalization was used in the older stages of Akkadian, it is notable that Neo-Assyrian, a relatively late dialect, presents us with two contrasting phenomena indicative of pharyngealization/velarization appearing as well as glottalization. The latter is indicated by the above-mentioned assimilation of a sibilant to *ʾāleph* to produce *ṣ* (see footnote 19), but the transformation of *qt* into *qṭ* in words such as *iqṭibi* (“he has said”, note 29) is more typical of “backing”. As mentioned, this is sometimes explained as a “group writing”; such a solution is, however, contraindicated by the fact that this type of writing appears even when the two consonants are separated by an epenthetic vowel, as in the writing *i-qi-ṭi-bu-ni-ṣū*.³⁵ Such a spelling would be extremely hard to explain as a “group writing” with the glottalic element moved to the end of a cluster, whereas it would fit eminently well with “backing”. This suggests that Neo-Assyrian was split on the issue of the realization of emphatics, with the older (glottalized/ejective) pronunciation appearing next to a “backed” one. One could speculate that this was due to influence from western languages,³⁶ perhaps specifically Aramaic. It is certainly interesting to note that the Neo-Assyrian phoneme showing the clearest evidence for glottalization is *ṣ* (see footnote 19), whereas *q* and *t* can be interpreted as having switched to “backing”/velarization/pharyngealization in that language: the emphatic affricate/sibilant *ṣ* was, after all, argued above to have kept its glottalization the longest in Northwest Semitic, and if areal influence from that linguistic

³⁴ Watson, 2009, 8–9.

³⁵ See Hämeen-Anttila, 2000, 21.

³⁶ Again, see Huehnergard, 1997, 438.

group really was the origin of the “backed” assimilation *qt* in Neo-Assyrian (as argued by Huehnergard), the data would fit perfectly, with precisely the phoneme *ṣ* apparently still being pronounced as an ejective in Neo-Assyrian despite the proposed areal influence from western languages.

Interdentals and sibilants in Ugaritic and Hebrew

Ugaritic *ṭ* as something like [s]

We now turn momentarily to non-emphatic phonemes. There is no real, hard evidence that Ugaritic *ṭ* was pronounced as an interdental: it certainly was so at an early Northwest Semitic period, which is suggested not only by the parallel sound in Arabic but also by Aramaic, which later shifted it to a dental occlusive *t*. In (the later stages of) Ugaritic, however, transcriptional material point to a sibilant realization (probably something close to [s]) – the *ṭ* letter is used to render sibilants of other languages, such as Hurrian and Hittite. The arguments adduced by Josef Tropper against this conclusion do not appear strong to me. In his view, the Ugaritians used *ṭ* for foreign sibilants because they did not (according to him) possess a simple [s] in their phonemic inventory (the phoneme transcribed as *s* represented an affricate [ts], at least at an early period) and thus used the most similar sound they did have for this purpose.³⁷ But one has to ask oneself: is an interdental fricative really that similar to an [s], when the palatal sibilant *š* is also available? Also, phonological systems completely lacking [s] are cross-linguistically rare (though not non-existent). The conclusion that Ugaritic *ṭ* was [s] (at least in later texts) is, to my mind, the most parsimonious one.

Deaffrication and interdentals

However, at some point, the originally affricate phonemes were apparently “deaffricated”.³⁸ I find it likely that the processes of deaffricating the phonemes *s* and *z* directly influenced the shift that appears to have occurred from interdental *ṭ* to a “neutral” sibilant and then further on to coalescing with the palatal sibilant *š* in some Northwest Semitic languages.

As a first stage of this process, the highly unstable interdental series lost its identity – a process we can see in action in Ugaritic, where *ṭ* appears to have come to be pronounced as something like [s], while **ḏ* often collapsed into the grapheme *d*, and the emphatic interdental *ẓ* (i. e. the emphatic interdental fricative) was also unstable, shown for example in the interesting *ḡ* merger men-

³⁷ Tropper, 2012, 112–113 (with references to earlier literature on the question). He has a similar argument concerning Ugaritic *ḏ* on p. 119.

³⁸ For the Ugaritic evidence for this, see Tropper, 2012, 103–104.

tioned above.³⁹ Indeed, in transcriptions of foreign (e. g. Hurrian) words and names, *ḏ* appears often to denote the voiced counterpart of [s], indicating that this letter did not necessarily signify an interdental sound in “Classical Ugaritic orthography” but rather something like a simple voiced sibilant [z] (a sound with which it did indeed become identified in Canaanite). This could mean that the historical *phoneme* **ḏ* normally developed into an ordinary *d* but that it in certain milieux became something like a phonetic [z], or otherwise (and perhaps more plausibly) that the few cases in which a voiced interdental fricative was kept in Ugaritic later had this sound further shifted into something like [z]. Perhaps the *ḏ* sign for etymological **ḏ* was something of an archaism and quite as important a use of the sign was actually transcribing foreign words/names? Or perhaps the sign was bivalent, covering both (a) the relatively uncommon actual etymological *ḏ* that had not shifted to *d* but was still pronounced interdentally and (b) the similar-sounding [z]-like phoneme of foreign words. In any case, the state of Ugaritic *ḏ* is yet another sign that the interdentals were unstable.

One example of the problems caused in the Ugaritic language by the collapsing state of the interdental series (and especially of its emphatic representative) is the very polymorphic word meaning something like “dream vision”. This word appears in Ugaritic in a number of different variants, namely *hḏrt*, *ḏhrt* and *ḏrt*. According to the view of Josef Tropper (with which I concur), we are dealing here (and in all three cases) with a relative of the Arabic root *zhr* (which has then been subjected to various forms of phonetic shifts).⁴⁰ It is quite notable that the Ugaritic reflex of this root does not show *z* at all. Rather, that unstable and apparently difficult sound has degenerated: it has lost its emphatic feature, and in the version *hḏrt* the interdental fricative has even degenerated into a (graphic) dental stop. This is yet another sign that the interdental series had started to become a ghost in later Ugaritic.

Appearance of new sibilant phonemes in Ugaritic

What seems to have happened after the interdental series had lost its identity is that the affricate series become deaffricated (excluding the emphatic *ṣ*, which, for reasons described above, probably kept its affricated pronunciation, at least in some dialects). This means that two new phonemes, [s] and [z], appeared. It is

³⁹ One should also mention that the Ugaritic grapheme *ḡ* may possibly in some uncertain cases reflect the outcomes of other emphatic fricatives as well – see Tropper, 2012, 95. This rather strengthens the argument of this article that the development of sibilants and sibilant-like sounds provided the impetus for “backing” becoming a common realization of emphasis in Northwest Semitic. See footnote 43 for further on these possible cases.

⁴⁰ See Tropper, 1996. One should note however, that Tropper believes that the emphatic interdental of Arabic *zhr* is probably secondary, and that the root originally contained a *ḏ*, as in Ugaritic.

of course not certain whether this “[s] from [ts]” phoneme was really identical to the “[s] from *t̥*” discussed earlier – probably it was not, for (as we shall see) their historical development was different. It is quite possible that there was some small articulatory difference between the two, the exact nature of which cannot be known.

My view is that, as a “new” unvoiced [s]-sound came into existence as a result of deaffrication, this provoked the “ex-interdental” [s] to shift into a palatal sound, coalescing with *ṣ*. This would, then, yield the situation found in the Canaanite languages. Such a model would create an explanation for the otherwise somewhat surprising development from an interdental fricative to a palatal sibilant.⁴¹

These developments can be illustrated in the following table (showing the process in Ugaritic in the first two stages; in the third stage, a sort of artificial conglomerate of a number of NWS processes is shown, without being intended to represent any one specific language).

Step I: early (pre-)Ugaritic, interdentals preserved as *such* and *s*-series affricate

<i>t̥</i> = [θ]	<i>ḏ</i> = [ð]	<i>ṣ</i> = [θʕ], [θʔ] or [θʕʕ]	
<i>s</i> = [ts]	<i>z</i> = [dz]	<i>ṣ</i> = [tsʕ], [tsʕʕ] or [tsʕʕʕ]	(glottalization of <i>ṣ</i> possibly preserved later than in the other phonemes)

Step II: interdentals collapse into other phonemes

<i>t̥</i> = [s]	<i>ḏ</i> > <i>d</i> (often)	<i>ṣ</i> > <i>ḡ</i> (sometimes)
<i>s</i> = [ts]	<i>z</i> = [dz]	<i>ṣ</i> = [tsʔ], [tsʕ] or [tsʕʕ]

Step III: *s*-series is mainly deaffricated, provoking further shifting of *t̥* to an *f*-sound (at a later dialectal stage)

<i>t̥</i> = [f]	<i>ḏ</i> > <i>d</i>	<i>ṣ</i> > <i>ḡ</i> (sometimes)	(<i>ḏ</i> and <i>ṣ</i> exemplified using Ug. development, not true for all NWS)
<i>s</i> = [s]	<i>z</i> = [z]	<i>ṣ</i> = [sʕ], [tsʕ] or [tsʕʕ]	(affrication dialectally preserved due to emphasis)

⁴¹ This type of argument is certainly not unique; it is similar to the push-chain analysis described (but not necessarily accepted) in Kogan, 2011, 70.

Possible conclusions from the interlocking developments of emphatics, sibilant and “quasi-sibilant” sounds in Northwest Semitic

The interdental and lateral series as provoking agents in the development of the emphatics

It is certainly highly interesting that two certain cases in which originally emphatic phonemes have come to be reflected as velar/uvular/pharyngeal letters (Ugaritic *ḡ* from **ḡ* and Aramaic *q* and *ʿayin* from the emphatic lateral fricative **ṣ*) occur in the interdental and lateral consonantal series (see argument number 1, above).⁴² I would say that it is quite thinkable that this is directly related to the beginning instability of these two series: when the two series started to lose their autonomy, just keeping their emphatic representatives as ejectives would pose a risk of them being absorbed into the originally affricate series (which is indeed what happened to both **ṣ* and **ḡ* in Hebrew, which shifted them to *ṣ*), due to the “ts-sounding” qualities produced by pronouncing an ejective with the primary articulation close to the teeth or alveolar wall. In order to keep the phonemic distinctiveness of these two sounds, Aramaic and Ugaritic may sometimes have changed the realization of the emphatic feature to some sort of “backing” in these cases.⁴³ Another (but similar) possibility is that it was the spuriously

⁴² One may note with some interest that there may be direct textual evidence for the transitional nature of Ugaritic *ḡ* in the pairing of the words *ḡlmt* and *ḡlmt* in KTU 1.3 VI 7–8 (in the part of the tablet previously known as KTU 1.8). A common (though not universal) interpretation is that both these words mean something like “darkness” and represent the word occurring in syllabic transcription as *ḡu-ul-ma-tu₄* (so, e. g., the *DUL*, *sub vocibus*; see also Wikander, 2014, 260; Tropper, 2012, 95, is skeptical of the connection. For the syllabic attestation, see Huehnergard, 1987, 98–100, 164–165, who, however, only identifies it with the alphabetic spelling *ḡlmt*. See also the bibliographical references in Huehnergard’s footnote 67). If the words are variants of the same lexeme or root, we would here have a cunning word-play upon a word with sounds in transition. This is, however, highly uncertain. Another word that may possibly point in the direction sketched in this article is the Ugaritic verb *mḡy* (“to come”). This verb is sometimes connected with the Common Semitic root *mḡz*, yet the third consonant of the root could defeat this connection (the possible relationship is mentioned, though not with great conviction, in Tropper, 2012, 95). If one, however presupposes that the Ugaritic interdental emphatic *ḡ* was a “problematic” and disintegrating sound in historically attested Ugaritic, one could imagine that its remnants of glottalization caused a dissimilatory process, turning the *ʾāleph* into a glide while the *ḡ* itself turned into the “backed” phoneme *ḡ*. This would, however, be rather an *ad hoc* solution to this phonological problem and could therefore be regarded as methodologically suspect. Thus, I simply mention it here without basing larger arguments upon it.

⁴³ Note, however, two much less certain but possible cases of etymological **ṣ* itself appearing as Ugaritic *ḡ* as well in Tropper, 2012, 95 (the words *ḡhw/y*, meaning “to lower

appearing “spontaneous and intrusive t” that came about due to glottalization of these phonemes that was dissimilated to some sort of backed sound or feature in order to escape identification with the affricate series, which led to (and was reinterpreted as) a “backing” of the entire sound (cf. the analysis mentioned in footnote 44). This could also be in line with a general lessening of affrication as a feature of Northwest Semitic languages.⁴⁴

oneself” and *gḥy*, “to burn, dry out”). The etymological uncertainty of these words makes me hesitate to view them as a defeater for the argument above about trying to avoid conflation with the affricate series. If these words really contain reflexes of *š that appear as Ugaritic ḡ, it may simply be a case of some form of analogical development to what was happening to the disintegrating interdental series. As pointed out by Tropper himself (p. 96, basing himself on Greenstein, 1998, 104 [*non vidi*]), these words provide a similar phonetic environment (containing resonants) to the ones showing ḡ from *z, which could possibly have provoked the similar development. It is possible that the presence of resonants helped bring about the change of z to ḡ as well, but the drive to keep phonological distinctiveness (see above) could still have been important. It is certainly interesting that Tropper (after Greenstein) mentions that the resonants may have created a sort of semi-lateralized phonetic milieu – the lateral emphatic fricative *ś does, after all, also change in various Northwest Semitic languages (and becomes written with “back” graphemes in Aramaic), so some similarity in the phonetic development could be imagined. Cf. with the typologically similar possible developments in Egyptian sketched above in footnote 15.

⁴⁴ The analysis of the history of *ś in Aramaic argued in Kogan, 2011, 99 (based on work by several other scholars) may in fact be very well combined with the model proposed here. The view represented in Kogan is that the Old Aramaic spelling with *q* represents something like [kxʔ] or [qxʔ], which would have come into being through a sort of backing dissimilation of an affricated, emphatic (= glottalized) lateral, that is, the original [tʰʔ] dissimilated into [kʰʔ] and finally turned into [kxʔ] or [qxʔ]. Later, so the theory goes, this complex cluster [kxʔ] or [qxʔ] would have been identified with [ɣ] (i.e. *ḡayin*, which was a separate phoneme in early Aramaic) and then, finally, with *ʿayin* (when Aramaic *ḡayin* collapsed into that phoneme). This interpretation, involved though it is, is alluring in many ways (for example, it fits well with transcriptional material from the Amarna onomastics). The idea that the Old Aramaic sound was pronounced something like [qxʔ] is, however, not at all incompatible with the analysis presented above: as mentioned in the main text, the appearance of some sort of velar or even uvular feature into the pronunciation of what was once *ś would in itself work as an introduction of a “backing” feature into the phonological system of the language, and could thus have constituted a starting point of an identification between the feature [EMPHATIC] and “backing” of some sort (i.e. velarization or perhaps pharyngealization or uvularization) – and the appearance of the “backed entity” ([qx] or [kx] in Kogan’s analysis) could very well have been because of the dissimilation of an “intrusive t”, as argued above. None of this causes problems for my suggestion that the demise of the lateral and interdental series helped provoke these changes.

Indeed, a fact which accords well with such a hypothesis is the phenomenon that the affected phoneme in Ugaritic was an interdental (a series which was demonstrably collapsing in that language) but not in Aramaic, which kept the interdentals separate in the early (Old Aramaic) period but later merged them not with sibilants/ex-affricates (as did Hebrew) but with dental stops.

If this line of reasoning is accepted, a startling possibility presents itself: that the appearance of pharyngealization/velarization/uvularization (or generally “backing”) as a possible realization of the emphatic feature in (Northwest) Semitic originally appeared in these particular phonemes (whereupon it may have spread to other parts of the phonological system). If this is so, the developments of the interdentals, sibilants, lateral fricatives and emphatics are not only parallel, but directly interlocking. A possible piece of evidence pointing in this direction comes from Aramaic: in Old Aramaic, the emphatic lateral fricative *š was (as mentioned) rendered using the letter *q*, which points to some type of backed articulation. However (and now we come to argument number 5 on our list of arguments earlier in the article), these early Aramaic texts also show “Geers’ law-like” dissimilation of two emphatics in the same word (e. g. *kyš*’ for “summer”, cf. Hebrew *qayiš* and Ugaritic *qz*),⁴⁵ which would fit better with the (older) glottalized pronunciation being current in the Aramaic of that time. As time wore on, later Aramaic kept some sort of backed realization of historical *š (using *ayin* to represent it graphically), but the phenomenon of dissimilation of emphatics is now gone, which may indicate that the glottalized/ejective pronunciation of the emphatics was not as current – indeed, the opposite process (emphasis spread) seems to be in action in the root *qtl* (versus earlier *qtl*, which appears to be the original form of the root given its cognates in Arabic and Ethiopic). In Classical Aramaic, the “summer” word appears as *qaytā*, with two emphatics in the same word not being regarded as problematic at all, which fits with the idea that a transition had occurred from a glottalized to a “backed” pronunciation of the Aramaic emphatics (at least in some dialects).

All this seems to strengthen the possibility that *š was one of the first phonemes to acquire a backed realization of the emphatic feature in Aramaic, whence it later spread into other parts of the phonological system.⁴⁶ One can

⁴⁵ See Greenspahn, 2002, 97. On the dissimilatory process not being present later, Zelmáněk (1996, 51) also mentions that Geers’ law was “no longer valid in younger Semitic languages”. There have been suggestions that the early Aramaic dissimilation of emphatics (which actually partly seems to persist into Mandaic) really represents a piece of substrate influence from Akkadian (cautiously mentioned as a possibility in Huehnergard, 2006, 8, n. 37), but this view was well argued against by Kaufman (1974, 121–122).

⁴⁶ The view of Bellem (2007, 152–156), that treats *š (= *d) as *not* having been an emphatic at all in Proto-Semitic (but holding that it was, however, “backed” in the sense of velarized or pharyngealized), is not the point that I am arguing here. I certainly think that

note with some interest that the Modern South Arabian languages show evidence of both glottalized and pharyngealized realizations of emphatics, which shows that the situation may very well have been in flux at earlier points in the history of the Semitic languages as well. As mentioned above in note 9, there also appears to be some limited Neo-Aramaic evidence for such a transitional stage.

An interlocking model

The type of analytical model presented here presupposes that the various developments discussed – the changing phonetic realization of the emphatics, the dissolution of the interdental triad, the deaffrication of affricate phonemes – should be studied not in isolation but as parts of interlocking and complex processes that made Northwest Semitic look the way it does in preserved texts and reading traditions. And the emphatics of ancient Northwest Semitic do not have to be analyzed as having been either “exactly like Arabic” or “transformed by Arabic influence”: development from glottalization to “backing” may have happened gradually, and by language-internal processes. It should, however, be pointed out that what has been presented here is a *model*, a possible explanation of what must have been a very complex series of interrelated changes in a number of closely related languages and dialects. Data from a number of languages have been adduced to illustrate and analyze a number of developmental trends, but there really is no other solution if one wants to study phonological and phonetic changes in early Northwest Semitic over a rather extended period of time.

The interrelations that have been proposed between the developments in Ugaritic and Hebrew/Canaanite imply that the borders between the various branches of Northwest Semitic were not absolute; rather, some of the phonetic changes appear to have developed and spread as areal features across the Levantine area. Indeed, the deaffrication of *s* and *z* has occurred in most branches of Semitic. The various sound-changes must not be studied in isolation, but as part as ongoing and mutually influencing processes.

the Proto-Semitic sound was emphatic (originally = glottalized/ejective) but that this articulation shifted in Aramaic, and that this shift (and the similar one in Ugaritic) provide clues as to how the pharyngealized/velarized realization of the emphatic feature came into being. Note that Bellem (p. 155) makes the same connection that I do between the backed pronunciation of *š (= *ḏ) and its Aramaic graphic representations as *q* and *ʿayin*, though with vastly different viewpoints. Bellem’s argument against *š (= *ḏ) having originally been emphatic (= ejective) appears mostly to be mediaeval accounts of the pronunciation of the phoneme in Arabic, which says nothing at all of its earlier realization. The Ethiopic languages show glottalization in their reflexes of the phoneme (though identified with *š*), which would be hard to explain unless it was, indeed, emphatic (= glottalized) from the beginning.

And, as we have seen, according to this model, these processes had a domino-like quality: the interdentals and laterals started collapsing into other sounds, *which* caused the inventory of *s*-like sounds to change, *while* the affricates started losing their affrication, *all of which together* caused or influenced backing of the emphatic interdental of Ugaritic and the emphatic lateral of Aramaic, *which* subsequently spread to the other emphatic phonemes. Interlocking, indeed.

Acknowledgment

An earlier version of this article was presented at the September 2014 OTSEM conference in Hamburg. I would like to thank the participants of the discussion for valuable remarks, especially Prof. Martti Nissinen, who acted as respondent for the paper.

References

- Aro, J., 1977: Pronunciation of the ‘Emphatic’ Consonants in Semitic Languages. *StOr* 47: 5–18.
- Bauer, Hans / Leander, Pontus, 1965 [1922]: *Historische Grammatik der hebräischen Sprache*. Olms Paperbacks 19. Hildesheim: Georg Olms Verlagsbuchhandlung.
- Bellem, Alex, 2007: *Towards a Comparative Typology of Emphatics: Across Arabic and into Arabic Dialect Phonology*. PhD thesis, Dept. of Linguistics, School of Oriental and African Studies, London University.
- Dolgopolsky, Aharon B., 1977: Emphatic Consonants in Semitic. *IOS* 7, 1–13.
- Edzard, Lutz, 1998: *Polygenesis, Convergence, and Entropy: An Alternative Model of Linguistic Evolution Applied to Semitic Linguistics*. Wiesbaden: Harrasowitz.
- Faber, Alice, 1980: *Genetic Subgroupings of the Semitic Languages*. PhD thesis, University of Texas at Austin.
- Greenspahn, Frederick E., 2002: Aramaic. In John Kaltner / Steven L. McKenzie (eds.): *Beyond Babel: A Handbook for Biblical Hebrew and Related Languages*. *Resources for Biblical Study* 42. Leiden/Boston, MA: Brill / Society of Biblical Literature. 93–108.
- Greenstein, E. L., 1998: On a New Grammar of Ugaritic. *IOS* 18, 397.
- Hackett, Jo Ann, 1984: *The Balaam Text from Deir ‘Allā*. Chico, CA: Scholars Press.
- 2010: *A Basic Introduction to Biblical Hebrew: with CD*. Peabody, MA: Hendrickson.
- Hämeen-Anttila, Jaakko, 2000: *A Sketch of Neo-Assyrian Grammar*. *SAAS* 13. Helsinki.

- Huehnergard, John, 1987: Ugaritic Vocabulary in Syllabic Transcription. HSS 32. Atlanta, GA: Scholars Press.
- 1997: Akkadian Grammar (rev. of von Soden / Mayer, Grundriss der akkadischen Grammatik, 3rd ed). Or 66, 434–444.
- 2006: Proto-Semitic and Proto-Akkadian. In G. Deutscher / N. J. C. Kouwenberg (eds.): The Akkadian Language in its Semitic Context: Studies in the Akkadian of the Third and Second Millennium BC. Uitgaven van het Nederlands Instituut voor het Nabije Oosten te Leiden 106. Leiden: Nederlands Instituut voor het Nabije Oosten. 1–23.
- Joüon, Paul / Muraoka, Takamitsu, 2006: A Grammar of Biblical Hebrew. 2nd, rev. ed. Subsidia Biblica 27. Editrice Pontificio Istituto: Roma.
- Kaufman, Steven A., 1974: The Akkadian Influence on Aramaic. The Oriental Institute of the University of Chicago. AS 19. Chicago/London: University of Chicago Press.
- 1997: Aramaic. In Robert Hetzron (ed.): The Semitic Languages.. Routledge Language Family Descriptions. London / New York, NY: Routledge. 114–130.
- Kogan, Leonid, 2011: 6. Proto-Semitic Phonetics and Phonology”. In Stefan Weniger et al. (eds.): The Semitic Languages: An International Handbook, Handbücher zu Sprach- und Kommunikationswissenschaft 36. Berlin/Boston: de Gruyter Mouton. 54–151.
- König, F. E., 1881, 1895: Historisch-Kritisches Lehrgebäude der hebräischen Sprache (vols. 1–2 of 3). Leipzig: J. C. Hinrich’sche Buchhandlung.
- Kouwenberg, N. J. C., 2003: Evidence for Post-Glottalized Consonants in Assyrian. JCS 55, 78–86.
- Leslau, Wolf, 1987: Comparative Dictionary of Ge’ez (Classical Ethiopic): Ge’ez-English, English-Ge’ez: With an Index of the Semitic Roots. Wiesbaden: Harrasowitz.
- Lipiński, Edward, 1997: Semitic Languages: Outline of a Comparative Grammar. OLA 80. Leuven: Peeters.
- Loewenstamm, Samuel E., 1978: אֲנִי אֶחָדָה. ZAW 90, 410.
- Loprieno, Antonio, 1995: Ancient Egyptian: A Linguistic Introduction. Cambridge: Cambridge University Press.
- Loretz, Oswald, 1975: Hebräisch *ḥwṭ* “bezahlen, erstatten” in Gen. 31,39. ZAW 87, 207–208.
- Peust, Carsten, 1999: Egyptian Phonology: An Introduction to the Phonology of a Dead Language. Monographien zur Ägyptischen Sprache 2. Göttingen: Peust & Gutschmidt Verlag.
- Rendsburg, Gary A., 1997: Chapter 5: Ancient Hebrew Phonology. In Alan S. Kaye (ed.): Phonologies of Asia and Africa: Including the Caucasus, vol. 1. Winona Lake, IN: Eisenbrauns. 65–83.

- Rössler, Otto, 1971: Das Ägyptische als Semitische Sprache. In Franz Altheim / Ruth Stiehl (eds.): *Christentum am Roten Meer*, vol. 1. Berlin. 263–326
- Simeone-Senelle, Marie-Claude, 1997: The Modern South Arabian Languages. In Robert Hetzron (ed.): *The Semitic Languages*. Routledge Language Family Descriptions. London / New York, NY: Routledge. 114–130.
- Steiner, Richard C., 1982: Affricated Ṣade in the Semitic Languages. *The American Academy for Jewish Research Monograph Series* 3. New York: Bet Sha'ar.
- Tropper, Josef, 1996: Ugaritic Dreams: Notes on Ugaritic *d(h)rt* and *hprt*. In N. Wyatt / W. G. E. Watson / J. B. Lloyd (eds.): *Ugarit, Religion and Culture: Proceedings of the International Colloquium on Ugarit, Religion and Culture, Edinburgh, July 1994: Essays Presented in Honour of Professor John C. L. Gibson*. UBL 12. Münster: Ugarit-Verlag. 305–313.
- 2012: *Ugaritische Grammatik*. 2nd ed. AOAT 273. Münster: Ugarit-Verlag.
- Vilenčik, J., 1933: Welchen Lautwert hatte ض im Ursemitischen? *OLZ* 33, 89–98.
- Watson, Janet C. E., 2009: Introduction. In Alexander Sima, *Mehri-Texte aus der jemenitischen Šarqīya* (with cooperation from 'Askari Ḥugayrān Sa'd, ed. J. C. E. Watson and W. Arnold). Wiesbaden: Harrasowitz. 1–28.
- Wikander, Ola, 2014: Drought, Death, and the Sun in Ugarit and Ancient Israel: A Philological and Comparative Study. *ConBOTS* 61. Winona Lake, IN: Eisenbrauns.
- Wilson-Wright, Aren, 2011: The Etymology of Biblical Hebrew נָחַם and Some Semitic Consonant Correspondences. (Available from www.academia.edu.)
- Zeláněk, Petr, 1996: *The Origins of Pharyngealization in Semitic*. Enigma Corporation: Praha.